

# When Null Does Not Mean Absent: Measurement-Map Geometry, Factorization Boundaries, and Null–Witness Duality

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A null experimental outcome is not, by itself, a statement of physical absence. A latent state is transformed through source physics, response formation, probe selection, ensemble averaging, instrument transfer, and nuisance structure before recorded data are produced. We formulate a measurement-map geometry that makes the resulting losses of distinguishability explicit while also stating the limits of stage-by-stage attribution. For a total measurement map  $\mathcal{M} : X \rightarrow Y$ , observationally equivalent states lie in the same fiber of  $\mathcal{M}$ , and a target quantity is identifiable exactly when it is constant on those fibers. In a declared layered linear experiment with cumulative maps  $F_j = A_j \cdots A_1$ , the cumulative null spaces  $K_j = \ker F_j$  form a filtration and the dual witness spaces  $W_j = \text{Ann}(K_j) = \text{im } F_j^*$  shrink in the opposite direction. We prove that this filtration is invariant under invertible reparameterizations of intermediate spaces but not under arbitrary regrouping or refactorization of the same total map; stage-local statements are therefore conditional physical-model claims, whereas the total observational quotient is factorization independent. We also distinguish passive downstream post-processing, which cannot restore a distinction already erased on the same data path, from active intervention or re-preparation, which creates a new measurement map and can reduce the joint null space. The quotient  $K_j/K_{j-1}$  records newly erased directions within the declared factorization, while  $d_C(A) = \dim C(\ker A)$  quantifies exact unresolved target directions without becoming an evidence score. Nonlinear, noisy, nuisance-limited experiments are treated through effective Jacobians and Fisher information. Stochastic experiments fit the same logic at the level of outcome distributions, and quantum measurements enter through the probability map induced by a specified POVM; sequential backaction requires the corresponding quantum instrument to be included explicitly. Altermagnetic neutron and multipole measurements provide motivating examples. The contribution is not new linear algebra; it is a compositional physics calculus with explicit invariance, intervention, and scope boundaries for stating what a measurement does and does not identify.

## I. INTRODUCTION

Physics often compresses an experiment into a binary statement: a signal is present or absent. The inference required to justify that statement is usually richer. A microscopic state or parameter set first determines a physical source state; that state generates response functions; a probe couples only to selected components; domains, orientations, or ensembles may be averaged; finite resolution and detector transfer further transform the signal; and nuisance structure enters before inference. The recorded datum therefore constrains the latent state only through a composite map.

The individual mathematical ingredients of this statement are established. Kalman observability identifies unobservable state directions in linear systems; structural identifiability asks whether distinct parameters can generate identical outputs; inverse-problem theory centers nonuniqueness, instability, and regularization; optimal experimental design compares measurements relative to an inferential objective; and information geometry describes weakly constrained directions of a model manifold.[1–9]

The practical problem is that these ideas are often invoked at different layers of a physics experiment without

a common bookkeeping language. A response that is physically zero, a response projected out by the probe, a domain-cancelled signal, an unresolved spectral feature, and a nuisance-confounded parameter can all produce superficially similar negative outcomes. Conversely, several positive measurements can repeat the same information and create the appearance of closure without eliminating the ambiguity relevant to the claim.

Recent altermagnetic experiments provide a useful motivating sequence because the same ordered state can be interrogated through electronic spin splitting, magnon energies, polarized-neutron chirality, real-space magnetic multipoles, and signed reversal channels.[14–20] These observables may be related by one physical model, but they are not observationally interchangeable. The point of the present paper is to abstract the common measurement geometry without turning any one material-specific relation into a universal law.

We therefore organize the analysis around three objects: the *measurement fibers* that encode observational equivalence, the *null filtration* that localizes where distinctions are destroyed, and the dual *witness spaces* that encode which target covectors remain measurable. The algebra is classical. The contribution is the layered composition and its use as a physics-facing discipline for null interpretation, cross-probe closure, and symmetry-coded tests.

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## II. LAYERED MEASUREMENT MAPS AND OBSERVATIONAL EQUIVALENCE

Let  $X$  denote a latent state or parameter space. We factor a generic experiment as

$$\begin{aligned} X &\xrightarrow{S} X_S \xrightarrow{\mathcal{R}} X_R \xrightarrow{\mathcal{P}} X_P \xrightarrow{\mathcal{A}} X_A \xrightarrow{\mathcal{T}} Y, \\ \mathcal{M} &= \mathcal{T} \circ \mathcal{A} \circ \mathcal{P} \circ \mathcal{R} \circ S. \end{aligned} \quad (1)$$

Here  $S$  encodes the source or Hamiltonian-to-state step relevant to the claim,  $\mathcal{R}$  produces the physical response,  $\mathcal{P}$  is probe selection or symmetry projection,  $\mathcal{A}$  represents ensemble, orientation, or domain averaging, and  $\mathcal{T}$  collects finite resolution, binning, detector transfer, and other downstream operations. Nuisance parameters may enter any stage and will be treated explicitly below.

## III. FACTORIZATION INVARIANCE, REFACTORIZATION, AND INTERVENTION

The total map  $\mathcal{M}$  determines observational equivalence, but it does not uniquely determine the physical narrative used to factor the experiment. This distinction is essential if a null is to be attributed to a particular stage rather than merely to the total measurement.

**Proposition 1** (Invariance under interstage isomorphisms). *Let a linear experiment be written as  $F_j = A_j A_{j-1} \cdots A_1 : X_0 \rightarrow X_j$ . Let  $L_j : X_j \rightarrow X'_j$  be invertible for  $j = 1, \dots, n-1$ , with  $L_0 = I$  and  $L_n = I$ , and define*

$$A'_j = L_j A_j L_{j-1}^{-1}. \quad (2)$$

*Then  $F'_j = L_j F_j$  and therefore*

$$\ker F'_j = \ker F_j \quad (3)$$

*for every cumulative stage. The null filtration is therefore invariant under invertible reparameterizations of the declared intermediate spaces.*

The proof follows immediately because multiplication by an invertible map on the codomain does not change the kernel. Equation (3) is the appropriate invariance statement for stage-local observability. It does *not* imply invariance under arbitrary changes in how the total experiment is partitioned into stages.

**Proposition 2** (Stage attribution is not a total-map invariant). *Consider*

$$\mathcal{M} = \begin{bmatrix} 1 & 0 \end{bmatrix} : \mathbb{R}^2 \rightarrow \mathbb{R}. \quad (4)$$

*The same total map admits the factorizations*

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad (5)$$

$$B_1 = I_2, \quad B_2 = \begin{bmatrix} 1 & 0 \end{bmatrix}. \quad (6)$$

*For the first factorization,  $\ker A_1 = \text{span}\{(0, 1)^T\}$ , so the  $x_2$  distinction is erased at stage 1. For the second,  $\ker B_1 = \{0\}$  and the same distinction is erased only at stage 2. Both give the identical total map  $\mathcal{M}$  and the identical total observational quotient.*

This counterexample fixes the interpretation of labels such as “response null” or “projection null”: they are meaningful only relative to a physically justified factorization. Arbitrary regrouping can move the apparent location of information loss even though it cannot alter the final measurement fibers.

**Proposition 3** (Passive data processing cannot restore erased distinctions). *Let  $\mathcal{M} : X \rightarrow Y$  be any deterministic measurement map and let  $B : Y \rightarrow Z$  be deterministic post-processing on the same recorded data stream. Then*

$$\mathcal{M}(x) = \mathcal{M}(x') \implies (B \circ \mathcal{M})(x) = (B \circ \mathcal{M})(x'). \quad (7)$$

*For linear maps this implies*

$$\ker \mathcal{M} \subseteq \ker(B\mathcal{M}). \quad (8)$$

*Thus passive downstream processing can preserve or destroy distinctions, but cannot recreate one that the upstream data no longer contain.*

Active intervention is different because it changes the experiment. For the map in Eq. (4), let  $U$  swap the latent coordinates,

$$U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad (9)$$

A second experiment after this controlled intervention measures  $\mathcal{M}U = \begin{bmatrix} 0 & 1 \end{bmatrix}$ . Stacking the original and intervened measurements gives

$$A_{\text{int}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \ker A_{\text{int}} = \{0\}. \quad (10)$$

The lost distinction has not been recovered by post-processing; a new measurement channel has been created. Field reversal, domain retraining, sample rotation, pump-probe preparation, and strain control belong to this intervention class whenever they alter the state or forward map before acquisition.

TABLE I. Explicit two-dimensional stress test. The total passive map is the same in the first two rows, but the stage assigned responsibility for the null changes. The final row is a new intervened experiment, not a refactorization or post-processing of the first datum.

| Case                          | first cumulative kernel | final kernel         |
|-------------------------------|-------------------------|----------------------|
| $A_2 A_1 = \mathcal{M}$       | $\text{span}\{e_2\}$    | $\text{span}\{e_2\}$ |
| $B_2 B_1 = \mathcal{M}$       | $\{0\}$                 | $\text{span}\{e_2\}$ |
| $[\mathcal{M}; \mathcal{M}U]$ | —                       | $\{0\}$              |

$$\mathcal{M} = \mathcal{T} \circ \mathcal{A} \circ \mathcal{P} \circ \mathcal{R} \circ \mathcal{S}$$

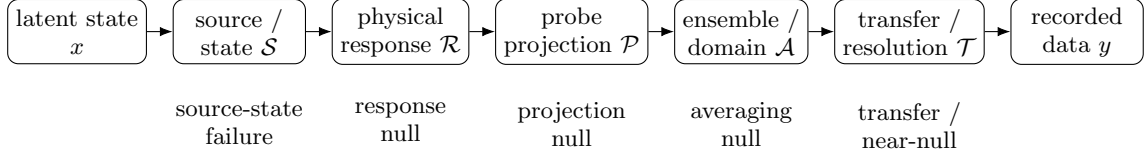


FIG. 1. A layered measurement map. Distinguishability can be lost at several physically different stages before data are recorded. The factorization is problem dependent, but it is useful only when the stages correspond to physically meaningful operations. Schematic TikZ code was developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the author specified the mathematical content and verified the final rendering.

**Definition 1** (Measurement fiber). *For a map  $\mathcal{M} : X \rightarrow Y$ , the measurement fiber through  $x \in X$  is*

$$F_{\mathcal{M}}(x) = \mathcal{M}^{-1}(\mathcal{M}(x)). \quad (11)$$

**Definition 2** (Observational equivalence). *Define*

$$x \sim_{\mathcal{M}} x' \iff \mathcal{M}(x) = \mathcal{M}(x'). \quad (12)$$

*The observable quotient is  $X_{\mathcal{M}} = X/\sim_{\mathcal{M}}$  with canonical projection  $\pi_{\mathcal{M}} : X \rightarrow X_{\mathcal{M}}$ .*

The quotient, rather than the original parameterization, is the object maximally determined by ideal data from  $\mathcal{M}$ . Global identifiability corresponds to singleton fibers; nonidentifiability corresponds to nontrivial fibers. This restates the classical logic of indistinguishable parameter values.[2, 3]

**Proposition 4** (Target factorization criterion). *Let  $f : X \rightarrow Z$  be a target quantity. The following are equivalent:*

1.  *$f$  is constant on every measurement fiber;*
2.  *$\mathcal{M}(x) = \mathcal{M}(x')$  implies  $f(x) = f(x')$ ;*
3. *there exists a unique map  $\bar{f} : X_{\mathcal{M}} \rightarrow Z$  such that*

$$f = \bar{f} \circ \pi_{\mathcal{M}}. \quad (13)$$

The proof is the universal property of the quotient by an equivalence relation. Proposition 4 shifts the experimental question from “is the entire model identifiable?” to “is the specific claim identifiable?” A model can contain nonidentifiable microscopic coordinates while a sign, tensor component, response coefficient, or prediction is fully determined.[9]

#### IV. THE OBSERVABILITY FILTRATION

Consider a finite-dimensional linear chain

$$V_0 \xrightarrow{A_1} V_1 \xrightarrow{A_2} \cdots \xrightarrow{A_n} V_n. \quad (14)$$

Define the cumulative map through stage  $j$  by

$$F_j = A_j A_{j-1} \cdots A_1, \quad F_0 = I, \quad (15)$$

and the cumulative null space by

$$K_j = \ker F_j, \quad K_0 = \{0\}. \quad (16)$$

**Proposition 5** (Null-space filtration). *The cumulative null spaces form a nested sequence*

$$K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n. \quad (17)$$

The proof is immediate:  $F_j v = 0$  implies  $F_{j+1} v = A_{j+1} F_j v = 0$ . No downstream deterministic operation can restore a distinction already erased upstream. Nonlinearly, the same statement holds for equivalence relations: if  $F_j(x) = F_j(x')$ , any deterministic post-processing  $B$  satisfies  $B(F_j(x)) = B(F_j(x'))$ .

The quotient

$$L_j = K_j / K_{j-1} \quad (18)$$

records equivalence directions that become newly invisible when stage  $j$  is appended. The associated graded object

$$\text{gr } K = \bigoplus_{j=1}^n K_j / K_{j-1} \quad (19)$$

is bookkeeping rather than a canonical decomposition of the original state space: it localizes where information is first lost relative to the chosen factorization.

Now let the claim be a linear target  $C : V_0 \rightarrow U$ .

**Proposition 6** (Target survival). *The target  $Cx$  is identifiable from the data after stage  $j$  if and only if*

$$K_j \subseteq \ker C. \quad (20)$$

Thus a target can be identifiable in an upstream response and become nonidentifiable after projection, averaging, or transfer. The first  $j$  for which Eq. (20) fails localizes the first irreversible target-relevant loss.

#### V. NULL-WITNESS DUALITY AND CONSTRUCTIVE OBSTRUCTIONS

The filtration has a dual description. Define the witness space

$$W_j = \text{Ann}(K_j) = \{\ell \in V_0^* : \ell(v) = 0 \ \forall v \in K_j\}. \quad (21)$$

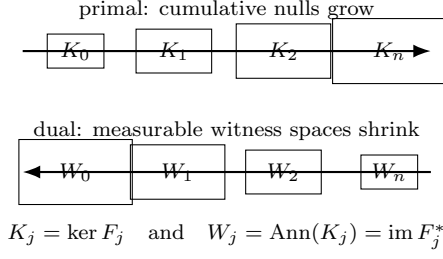


FIG. 2. Null-witness duality. Downstream composition can only enlarge the cumulative null space while the measurable covector space contracts. The two filtrations encode the same loss of distinguishability from primal and dual viewpoints. Schematic TikZ code was developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the author specified and verified the mathematical content.

In finite-dimensional inner-product spaces,

$$W_j = \text{im } F_j^*, \quad (22)$$

so the dual sequence runs in the opposite direction,

$$W_0 \supseteq W_1 \supseteq \cdots \supseteq W_n. \quad (23)$$

A target covector survives exactly while it remains in the witness space available at the data stage.

The duality also supplies a falsifier. If there exists

$$v \in K_n \quad \text{with} \quad Cv \neq 0, \quad (24)$$

then two states separated by  $v$  are experimentally indistinguishable but imply different target values. Equation (24) is a constructive obstruction to the inference.

For a multidimensional target, define the *target-obstruction dimension*

$$d_C(A) = \dim C(\ker A). \quad (25)$$

This is zero exactly when  $Cx$  is identifiable from  $Ax$ . It is neither an evidence score nor a probability of truth; it counts exact target dimensions left unresolved by the stated linear map.

## VI. INDEPENDENT PROBES, CLOSURE, AND STATE CONSISTENCY

Suppose independent probes have linear maps  $A_i : V \rightarrow W_i$ . The stacked experiment

$$A_{\text{stack}} = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_m \end{bmatrix} \quad (26)$$

satisfies

$$\ker A_{\text{stack}} = \bigcap_{i=1}^m \ker A_i. \quad (27)$$

Dually,

$$\text{im } A_{\text{stack}}^* = \sum_{i=1}^m \text{im } A_i^*. \quad (28)$$

Therefore target closure is exact when

$$\bigcap_i \ker A_i \subseteq \ker C. \quad (29)$$

Complete reconstruction of every latent coordinate is unnecessary. Complementarity is target specific: an additional probe is useful only if it removes null directions that change the claim.

This intersection rule carries a physical prerequisite that is easy to miss. The probes must refer to a common latent state. If measurement  $i$  depends on uncontrolled state variables  $\zeta_i$ —for example strain, domain population, stoichiometry, temperature, surface reconstruction, or preparation history—then the correct model is

$$y_i = \mathcal{M}_i(x, \zeta_i), \quad (30)$$

not  $y_i = \mathcal{M}_i(x)$  with the  $\zeta_i$  silently ignored. If the  $\zeta_i$  differ and are not modeled, a formal kernel intersection can create spurious closure. Cross-probe closure therefore requires either common-state measurements or an enlarged latent space containing the relevant state variables.

**Corollary 1** (Monotonic target closure under exact common-state probes). *For exact measurements on a common latent state, adding a probe cannot increase  $d_C$ .*

The qualifier “common-state” is essential. It separates genuine information gain from comparison of physically different systems.

## VII. NONLINEAR, NOISY, AND NUISANCE-LIMITED EXPERIMENTS

Exact kernels describe structural identifiability. Real experiments are nonlinear and noisy. Let  $\theta$  denote parameters and  $m(\theta)$  model predictions. Around  $\theta_0$ , define the sensitivity matrix

$$J = \left. \frac{\partial m}{\partial \theta} \right|_{\theta_0}. \quad (31)$$

If the measurement covariance is  $\Sigma$  and nuisance directions span the columns of  $B$ , whiten the data space and project away the nuisance tangent space. Let  $Q_B$  be the orthogonal projector onto  $\text{im}(\Sigma^{-1/2}B)$ . The nuisance-orthogonal sensitivity is

$$J_{\text{eff}} = (I - Q_B)\Sigma^{-1/2}J. \quad (32)$$

The corresponding local information matrix is

$$F_{\text{eff}} = J_{\text{eff}}^T J_{\text{eff}}. \quad (33)$$

Hence

$$\ker F_{\text{eff}} = \ker J_{\text{eff}}. \quad (34)$$

Zero singular values signal local structural nonidentifiability. Small but nonzero singular values signal weak practical identifiability relative to the adopted noise and nuisance model.[3, 8, 9]

For a chosen tolerance  $\epsilon$ , one may define an experiment-specific practical near-null subspace

$$\mathcal{N}_\epsilon = \text{span}\{v_k : \sigma_k(J_{\text{eff}}) \leq \epsilon\}. \quad (35)$$

The threshold  $\epsilon$  is not universal; it depends on precision, model discrepancy, calibration stability, and the decision being made. Finite resolution acts through this local geometry because convolution, binning, or bandwidth restrictions alter the singular spectrum of the composed Jacobian.

A prior or regularizer changes a different object. It can select a representative from a broad likelihood fiber or stabilize a near-null direction, but it does not retroactively make the data map injective. Bayesian inference legitimately combines likelihood and prior information; a data-identifiability claim should keep their contributions distinct.[4, 5, 10]

### VIII. STOCHASTIC AND QUANTUM MEASUREMENT SCOPE

The deterministic map notation is most natural for expectation values, fitted observables, and noiseless forward models. The fiber logic extends to stochastic measurements by replacing a deterministic output with an outcome distribution. Let  $K(dy | x)$  be a Markov kernel. Define

$$x \sim_K x' \iff K(\cdot | x) = K(\cdot | x'). \quad (36)$$

A target is identifiable from the stochastic experiment only if it is constant on these distributional equivalence classes. If a second Markov kernel  $L(dz | y)$  passively post-processes the outcome, then equality of the  $K$ -distributions implies equality of the composed  $LK$ -distributions. The same no-recovery principle as Eq. (7) therefore holds at the level of probability laws.

Quantum measurements fit this statement once the measurement itself is specified. For a density operator  $\rho$  and POVM  $\{E_y\}$ ,

$$p(y | \rho) = \text{Tr}(\rho E_y) \quad (37)$$

defines the relevant outcome-distribution map. Two quantum states are observationally equivalent for that experiment when they induce the same probabilities for every recorded outcome. Noncommuting POVMs are different measurement maps and can reduce the joint equivalence class when performed on appropriate independently prepared systems. Sequential measurement backaction,

adaptive control, and state disturbance require the corresponding quantum instrument or channel to be included as part of the forward model; they are not represented by passive classical post-processing alone.[12, 13]

Accordingly, “measurement-agnostic” in this paper means agnostic across classical deterministic forward models and specified outcome-distribution maps, including POVM-induced quantum probabilities. It does not mean that arbitrary quantum measurement dynamics can be ignored.

### IX. SYMMETRY REVERSALS AS EXPERIMENTAL CODING

Controlled transformations—polarization reversal, field reversal, domain reversal, momentum-partner exchange, sample rotation, or strain sign—can be used as experimental coding operations. For a finite group  $G$  acting on measurement settings, standard representation projectors decompose the response into symmetry sectors.[11]

For  $N$  binary control labels  $s_i \in \{\pm 1\}$ , the Walsh–Hadamard coefficients below are exact factorial contrasts for the complete labeled hypercube. Their stronger interpretation as characters of the physical group  $G = (\mathbb{Z}_2)^N$  additionally requires the realized reversals to behave as independent involutions on the prepared state:  $r_i^2 \simeq I$  and  $r_i r_j \simeq r_j r_i$  within experimental tolerance. Hysteresis or path-dependent state preparation can violate that interpretation even though the contrast algebra remains valid. Under the validated group interpretation, the character coefficients are

$$\hat{I}_A = \frac{1}{2^N} \sum_{\mathbf{s}} \left( \prod_{i \in A} s_i \right) I(\mathbf{s}). \quad (38)$$

Equation (38) is a Walsh–Hadamard transform of the reversal cube. Contributions with different parity characters occupy orthogonal coefficients. If a model forbids a sector, the corresponding coefficient is a built-in obstruction channel: a significant nonzero value can indicate leakage, uncontrolled nuisance structure, or missing physics.

This is the symmetry-coded counterpart of Eq. (24). In the primal picture, one searches for a state displacement that leaves the data invariant while changing the target. In the dual picture, one constructs a measurement character that should vanish unless the target or a confounder occupies that sector. The two are not separate philosophies; they are complementary implementations of target-specific falsification.

### X. A COMPOSITIONAL TAXONOMY OF NULLS

The word “null” should identify the target that is not determined and the stage at which distinguishability is lost. Table II extends the four-null distinction that arises

naturally in polarized neutron scattering into a general layered taxonomy.

The taxonomy separates two questions routinely conflated in practice: “was a nonzero detector signal observed?” and “does this experiment identify the claimed physical quantity?” A large raw signal can be nonidentifying, and a nominally null scalar channel can coexist with a nonzero signal in an orthogonal symmetry sector.

## XI. WORKED MEASUREMENT PATTERNS

### A. Sum, difference, and polarization

Consider  $x = (a, b)^T$  and an unselective probe

$$A_+ = \begin{bmatrix} 1 & 1 \end{bmatrix}. \quad (39)$$

Its null space is spanned by  $(1, -1)^T$ , so the difference  $a - b$  is not identifiable. A second probe

$$A_- = \begin{bmatrix} 1 & -1 \end{bmatrix} \quad (40)$$

intersects the original null space only at zero. Polarization reversal, balanced detection, and many signed-control experiments implement this algebra physically. The difference channel is not revealed because its underlying coefficient has become larger; it is revealed because a new witness covector has been added.

This is directly relevant to polarized neutron scattering. The unpolarized contraction of a transverse magnetic correlation tensor and the polarization-odd antisymmetric channel are different projections. A null in the latter cannot be inferred from the former without an explicit factorization relation.[16, 18]

### B. Finite resolution

Let  $s(\omega; \theta)$  be an ideal spectrum and let the instrument record

$$y(\omega) = \int d\omega' R(\omega - \omega') s(\omega'; \theta). \quad (41)$$

Even if the ideal map  $\theta \mapsto s$  is injective, convolution can drive singular directions of the composed Jacobian below practical resolution. A resolution-limited null therefore need not be a physical null. The relevant object is the singular spectrum of the complete transfer-composed sensitivity, not the intrinsic line spacing alone. The contrast between high-resolution unpolarized and polarization-resolved measurements in  $\text{MnF}_2$  provides a concrete correlated-matter example of this distinction.[17, 18]

### C. Same total map, different null location

The matrices in Eqs. (5)–(6) are deliberately elementary because they isolate a conceptual point that can be

obscured in realistic spectroscopy. In both factorizations the experiment records only  $x_1$  and therefore has the same observational equivalence  $x \sim x'$  whenever  $x_1 = x'_1$ . Yet the first bookkeeping choice attributes loss of  $x_2$  to the first stage, whereas the second attributes it to the final stage. The physically defensible attribution must therefore come from an independently specified model of what the stages mean, not from the total input-output relation alone.

The intervention example in Eq. (10) supplies the dual lesson. If a controlled operation rotates or re-prepares the state before a second acquisition, the stacked experiment may identify a direction invisible to either acquisition alone. Such gain is real, but it is gain from *experiment diversification*, not from computational recovery of information absent in the first data stream.

### D. Static rank versus dynamical chirality

Polarized inelastic neutron scattering can measure a dynamical antisymmetric correlation channel, while polarized diffraction can reconstruct static magnetization density and its multipolar decomposition.[18, 19] Symmetry may relate these objects, but they are generated by different forward maps. In the present language, a static multipole-rank functional  $f_K$  is identifiable from a dynamical chiral measurement  $\mathcal{M}_\chi$  only if

$$f_K = \bar{f}_K \circ \pi_{\mathcal{M}_\chi}. \quad (42)$$

Symmetry compatibility alone does not establish this factorization. This is a general prohibition against claim migration across distinct response tensors.

### E. Source coordinates and readout gain

A small Hamiltonian coordinate can be the leading source of a selected response without being amplified in the Hamiltonian norm. If  $\lambda$  is odd under a reference symmetry and a response satisfies

$$R_-(\lambda) = \chi_1 \lambda + O(\lambda^3), \quad (43)$$

then a control variable can increase the measurement derivative  $|\partial O / \partial \lambda|$  without changing  $|\lambda|$  itself. Field-based proposals for the weak seventh-neighbor exchange imbalance in  $\text{MnF}_2$  illustrate this distinction between source magnitude and transduction gain.[20] In measurement-map language, the experiment changes the witness geometry, not the microscopic coefficient.

## XII. RELATION TO ESTABLISHED THEORY AND NOVELTY BOUNDARY

Several statements above are standard mathematics in familiar clothing. Quotients by kernels, annihilators,

TABLE II. Nulls and failure modes in a layered measurement map. A “null” means non-identifiability or zero sensitivity of a specified target, not necessarily zero raw detector counts.

| Class                    | Mathematical signature                                                       | Physical interpretation                                                                                            |
|--------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Source/state failure     | relevant source coordinate or prerequisite state absent before $\mathcal{R}$ | the hypothesized physical source state is not realized in the stated material/condition                            |
| Response null            | target variation already lies in a fiber of $\mathcal{R} \circ \mathcal{S}$  | the source exists, but the chosen response tensor or function vanishes by symmetry, matrix element, or dynamics    |
| Projection null          | response difference lies in $\ker \mathcal{P}$                               | the probe is insensitive to the relevant response component                                                        |
| Averaging/domain null    | symmetry or ensemble average projects the target sector to zero              | opposite domains, orientations, or populations cancel in the measured average                                      |
| Transfer/resolution null | projected difference lies in $\ker \mathcal{T}$                              | finite bandwidth, convolution, integration, binning, or detector response erases distinguishability                |
| Nuisance confounding     | target tangent lies in the nuisance tangent span                             | a reproducible signal cannot be uniquely separated from calibration, background, leakage, or preparation variables |
| Practical near-null      | exact sensitivity is nonzero but lies in a small singular-value direction    | structurally identifiable information is too weak for the available precision or model accuracy                    |
| Model inconsistency      | no common latent state reproduces independent data                           | not a measurement null: the stated source model or cross-probe state hypothesis fails                              |

adjoint images, and finite-group projectors are classical. Kalman observability and structural identifiability already formalize indistinguishable state or parameter directions. Fisher-information rank is a local identifiability diagnostic under regularity assumptions, and optimal-design theory already compares experiments relative to an inferential objective.[1, 3, 6, 7, 11]

The contribution here is therefore not a replacement theorem. It is a compositional synthesis with five concrete additions to physics-facing practice. First, the source–response–projection–averaging–transfer factorization localizes physically different losses of distinguishability. Second, the cumulative null filtration  $K_j$  and dual witness filtration  $W_j$  provide stage-by-stage bookkeeping, with  $K_j/K_{j-1}$  recording newly introduced equivalence directions. Third,  $d_C(A) = \dim C(\ker A)$  keeps closure target specific rather than global. Fourth, nuisance projection, practical conditioning, and finite-group reversal channels are placed in the same geometry. Fifth, multi-probe closure is conditioned on a common latent material state, preventing independent-looking data from being combined across uncontrolled changes of sample state.

The framework has equally important limits. Global nonlinear quotient spaces can be singular, disconnected, or inconvenient; local Jacobian rank can miss discrete global degeneracies; Fisher information depends on the likelihood model and does not prove global structural identifiability. More importantly, the factorization in Eq. (1) is not unique. The total observational quotient and any target factorization through that quotient are invariant properties of the specified total experiment. Stage-local attribution is invariant under invertible reparameterizations of the declared intermediate spaces [Eq. (3)] but is not invariant under arbitrary refactorization, as Eqs. (5)–(6) demonstrate. A label such as “projection null” is there-

fore a causal/physical modeling hypothesis that must be justified independently of the algebra. In addition, the no-recovery statement applies to passive same-path post-processing; an intervention that changes the state or forward map creates a new experiment and can reduce the joint null space [Eq. (10)]. Finally, the deterministic notation is not a claim that measurement outcomes are fundamentally deterministic. Stochastic and quantum experiments are included only after their probability map or measurement instrument has been declared explicitly. The taxonomy therefore identifies where a *stated* forward model loses target-relevant distinguishability; it does not by itself establish causality or model completeness.

### XIII. DISCUSSION

The central object of an experiment is often not a unique microscopic parameter vector but the equivalence class of states that the experiment cannot distinguish. This view makes negative results more local. A source-state failure is not a projection null; a domain cancellation is not a resolution limit; and a nuisance-confounded positive signal is not the same as a closed target. It also makes positive results more precise: a target can be fully identified even while many microscopic coordinates remain unconstrained.

The factorization boundary adds an equally important discipline. The safest invariant statement concerns the total fibers of the declared experiment. Stage-local explanations require more: the intermediate objects must correspond to independently defensible physical operations, and alternative physically plausible factorizations should be checked when attribution matters. This makes the framework intentionally asymmetric: total nonidenti-

fiability can be established from the total map, whereas a causal label for *where* information was lost is conditional on the forward-model architecture.

The null–witness duality gives a useful discipline for designing and interpreting measurements. On the primal side, downstream composition enlarges indistinguishable state directions. On the dual side, the available witness covectors shrink. A new probe is valuable when it adds a witness direction absent from the existing experiment, or equivalently when it removes a target-relevant null direction. A reversal channel is valuable when it places the target in a parity sector separated from dominant backgrounds and simultaneously creates forbidden sectors that can expose leakage or missing physics.

The cross-material lesson is equally important. A measurement-map framework should be allowed to stop upstream. If the prerequisite material state is contradicted, downstream signatures cannot repair that source-state hypothesis. Conversely, a downstream null need not invalidate the source if the null is demonstrably introduced by projection, averaging, or transfer. This asymmetry is one reason that a scalar “evidence score” is a poor substitute for a layered claim structure.

#### XIV. CONCLUSION

For a measurement map  $\mathcal{M} : X \rightarrow Y$ , the maximally data-identifiable object is the observational quotient

$$X_{\mathcal{M}} = X / \sim_{\mathcal{M}}, \quad (44)$$

and a target is identifiable exactly when it factors through that quotient. In layered linear experiments,

$$K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n, \quad W_0 \supseteq W_1 \supseteq \cdots \supseteq W_n, \quad (45)$$

with  $K_j = \ker F_j$  and  $W_j = \text{Ann}(K_j) = \text{im } F_j^*$ . Independent exact probes obey

$$\ker A_{\text{stack}} = \bigcap_i \ker A_i, \quad (46)$$

provided they refer to a common latent state. Nonlinear and noisy problems replace exact kernels with local effective geometry encoded by nuisance-projected Jacobians and Fisher information, while reversal experiments implement symmetry projectors that create both target witnesses and explicit obstruction channels.

Two boundaries accompany these statements. First, the total measurement fibers are invariant, whereas stage-local null attribution is only invariant under invertible

reparameterization of a physically declared factorization and not under arbitrary refactorization. Second, passive downstream processing cannot restore a distinction absent from the same data path, while an active intervention creates a new measurement map whose joint kernel can be smaller. For stochastic or quantum measurements, these claims apply to the declared outcome-distribution map or instrument.

The resulting rule is compact:

|                                                                                                                                                                |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| a null is local to a measurement map,<br>not automatically to the physics;<br>a positive signal identifies only what is constant<br>on its measurement fibers. | (47) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|

A physical system can contain more structure than any one experiment can identify. Scientific inference is sharper when the lost structure is represented explicitly rather than silently assigned a value.

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#### DATA AVAILABILITY

No new experimental data were generated for this theoretical and methodological work.

#### COMPETING INTERESTS

The author declares no competing interests.

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[1] R. E. Kalman, “Mathematical Description of Linear Dynamical Systems,” *J. Soc. Ind. Appl. Math. Ser. A Control* **1**, 152–192 (1963), doi:10.1137/0301010.

[2] R. Bellman and K. J. Åström, “On Structural Identifiability,” *Math. Biosci.* **7**, 329–339 (1970), doi:10.1016/0025-5564(70)90132-X.

- [3] T. J. Rothenberg, “Identification in Parametric Models,” *Econometrica* **39**, 577–591 (1971), doi:10.2307/1913267.
- [4] A. Tarantola, *Inverse Problem Theory and Methods for Model Parameter Estimation* (SIAM, Philadelphia, 2005), doi:10.1137/1.9780898717921.
- [5] J. Kaipio and E. Somersalo, *Statistical and Computational Inverse Problems* (Springer, New York, 2005), doi:10.1007/b138659.
- [6] K. Chaloner and I. Verdinelli, “Bayesian Experimental Design: A Review,” *Statistical Science* **10**, 273–304 (1995), doi:10.1214/ss/1177009939.
- [7] F. Pukelsheim, *Optimal Design of Experiments* (SIAM, Philadelphia, 2006), doi:10.1137/1.9780898719109.
- [8] M. K. Transtrum, B. B. Machta, and J. P. Sethna, “Geometry of Nonlinear Least Squares with Applications to Sloppy Models and Optimization,” *Phys. Rev. E* **83**, 036701 (2011), doi:10.1103/PhysRevE.83.036701.
- [9] A. Raue *et al.*, “Structural and Practical Identifiability Analysis of Partially Observed Dynamical Models by Exploiting the Profile Likelihood,” *Bioinformatics* **25**, 1923–1929 (2009), doi:10.1093/bioinformatics/btp358.
- [10] A. M. Stuart, “Inverse Problems: A Bayesian Perspective,” *Acta Numerica* **19**, 451–559 (2010), doi:10.1017/S0962492910000061.
- [11] J.-P. Serre, *Linear Representations of Finite Groups* (Springer, New York, 1977), doi:10.1007/978-1-4684-9458-7.
- [12] A. S. Holevo, *Probabilistic and Statistical Aspects of Quantum Theory* (North-Holland, Amsterdam, 1982).
- [13] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th Anniversary ed. (Cambridge University Press, Cambridge, 2010), doi:10.1017/CBO9780511976667.
- [14] L. Šmejkal, J. Sinova, and T. Jungwirth, “Emerging Research Landscape of Altermagnetism,” *Phys. Rev. X* **12**, 040501 (2022), doi:10.1103/PhysRevX.12.040501.
- [15] T. Jungwirth *et al.*, “Symmetry, Microscopy and Spectroscopy Signatures of Altermagnetism,” *Nature* **649**, 837–847 (2026), doi:10.1038/s41586-025-09883-2.
- [16] P. A. McClarty, A. Gukasov, and J. G. Rau, “Observing Altermagnetism Using Polarized Neutrons,” *Phys. Rev. B* **111**, L060405 (2025), doi:10.1103/PhysRevB.111.L060405.
- [17] V. C. Morano, Z. Maesen, S. E. Nikitin, J. Lass, D. G. Mazzone, and O. Zaharko, “Absence of Altermagnetic Magnon Band Splitting in  $\text{MnF}_2$ ,” *Phys. Rev. Lett.* **134**, 226702 (2025), doi:10.1103/PhysRevLett.134.226702.
- [18] Q. Faure *et al.*, “Altermagnetism Revealed by Polarized Neutrons in  $\text{MnF}_2$ ,” arXiv:2509.07087 (2025), to appear in *Science*, doi:10.48550/arXiv.2509.07087.
- [19] I. Kibalin *et al.*, “Real-Space Manifestation of Ferroic Multipoles in Altermagnetic  $\text{MnF}_2$ ,” arXiv:2608.29089 (2026), doi:10.48550/arXiv.2608.29089.
- [20] G. Jiang *et al.*, “Field-Amplified Readouts of Weak Altermagnetic Exchange in  $\text{MnF}_2$ ,” arXiv:2608.26540v2 (2026), doi:10.48550/arXiv.2608.26540.